# Sources of Magnetic Field

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# Learning Outcomes

**In this chapter, you'll learn…**

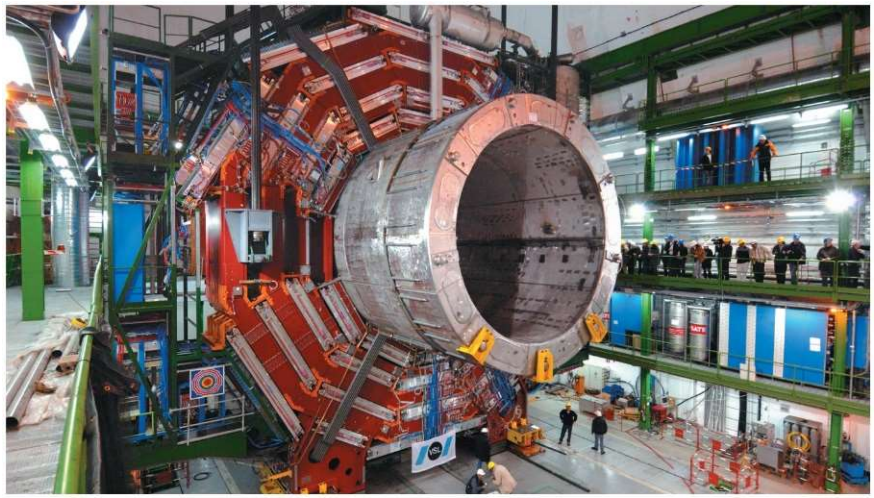
- how to calculate the magnetic field produced by a single moving charged particle, a straight current-carrying wire, or a current-carrying wire bent into a circle.
- why wires carrying current in the same direction attract, while wires carrying opposing currents repel.
- what Ampere's law is, and how to use it to calculate the magnetic field of symmetric current distributions.
- how microscopic currents within materials give them their magnetic properties.

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# Introduction

- The immense cylinder in this photograph is a current-carrying coil, or solenoid, that generates a uniform magnetic field in its interior as part of an experiment at CERN, the European Organization for Nuclear Research.
- What can we say about the magnetic field due to a solenoid?
- What actually creates magnetic fields?
- We will introduce Ampere's law to calculate magnetic fields.

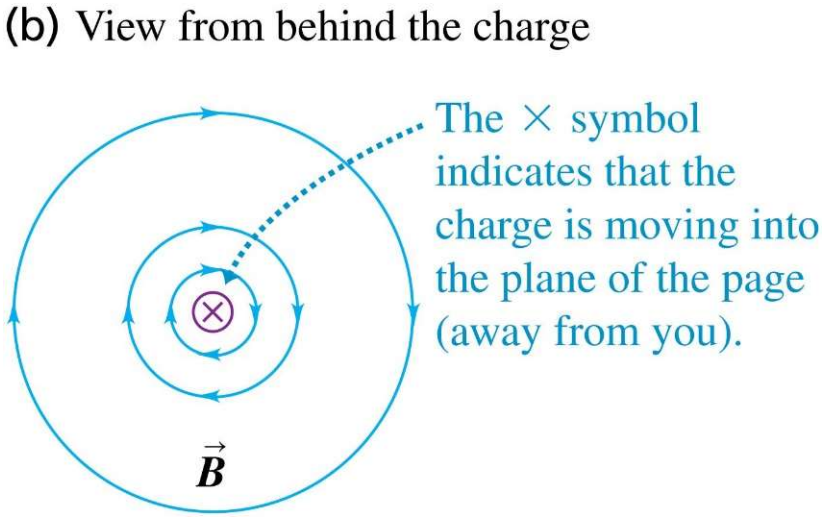


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# The Magnetic Field of a Moving Charge

- A moving charge generates a magnetic field that depends on the velocity of the charge, and the distance from the charge.




Magnetic field due to a point charge with constant velocity

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}$$

Magnetic constant · Charge · Velocity · Unit vector from point charge toward where field is measured · Distance from point charge to where field is measured

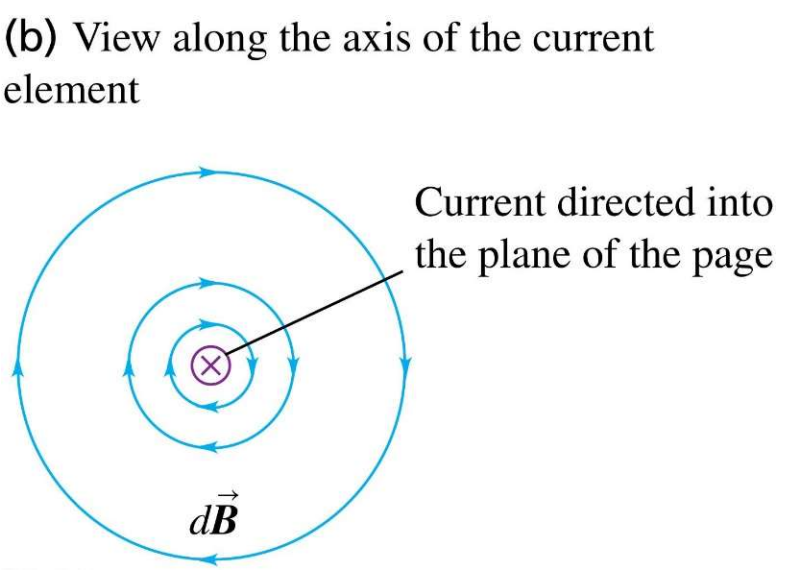
- Video Tutor Solution: Example 28.1

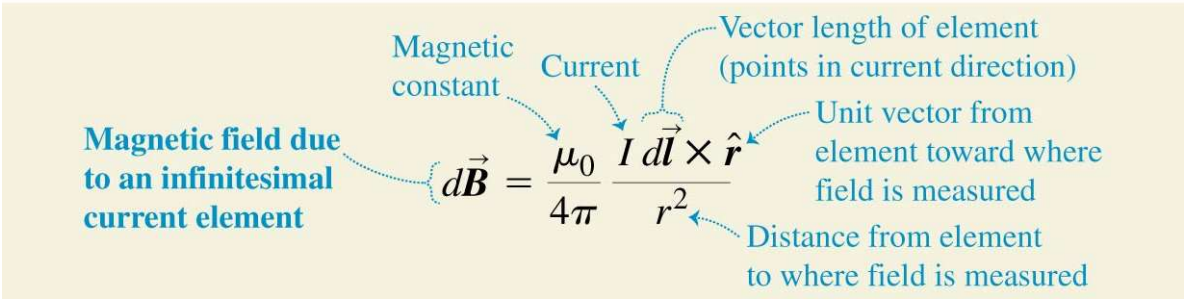
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# Magnetic Field of a Current Element

- The total magnetic field of several moving charges is the vector sum of each field.
- The magnetic field caused by a short segment of a current-carrying conductor is found using the **law of Biot and Savart**:



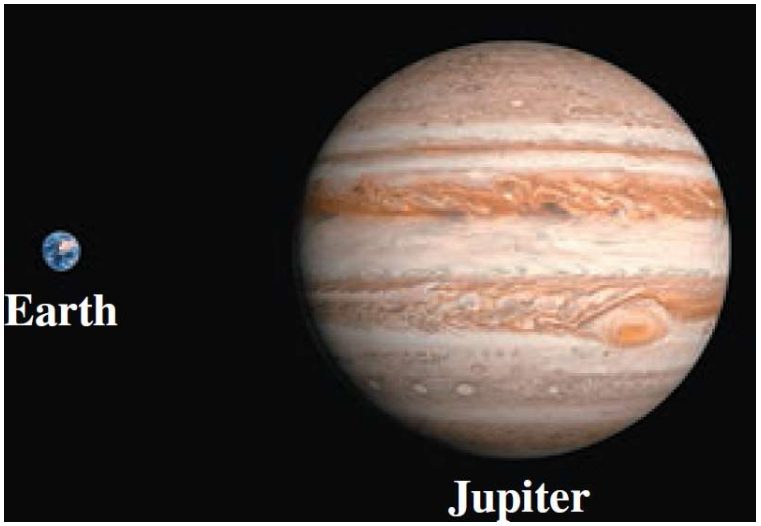




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# Currents and Planetary Magnetism

- The earth's magnetic field is caused by currents circulating within its molten, conducting interior.
- These currents are stirred by our planet's relatively rapid spin (one rotation per 24 hours).
- The planet Jupiter's internal currents are much stronger: Jupiter is much larger than the earth, has an interior that is mostly liquid hydrogen, and spins very rapidly.
- Hence Jupiter's magnetic field is about $2 \times 10^4$ greater than the earth's field.

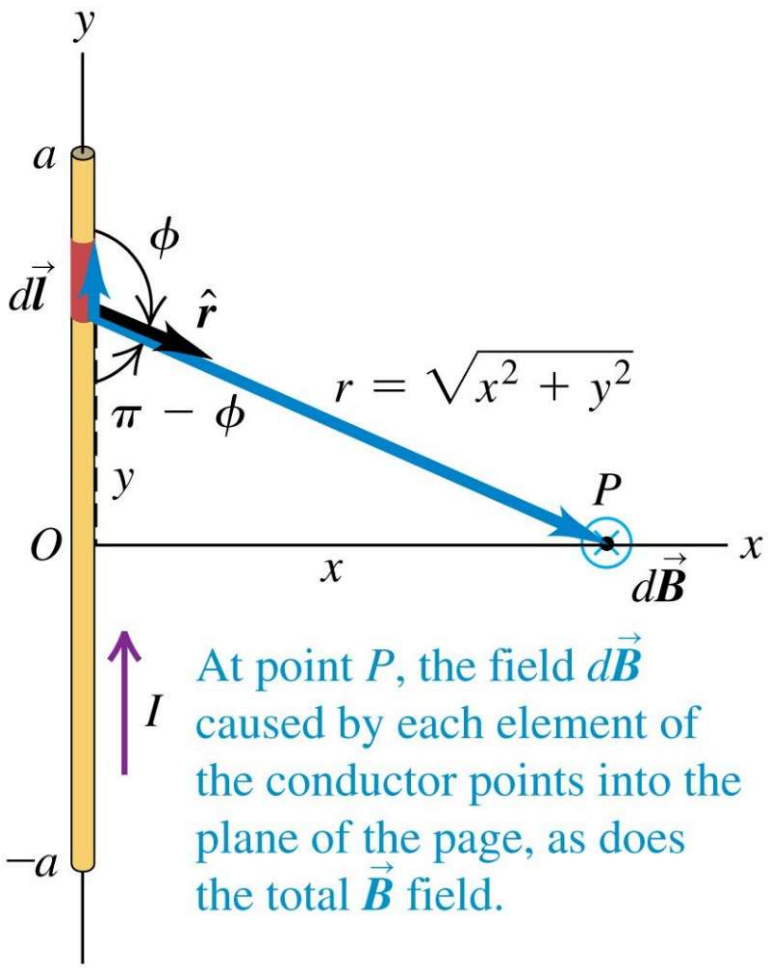



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# Magnetic Field of a Straight Current-Carrying Conductor (1 of 3)

- Let's use the law of Biot and Savart to find the magnetic field produced by a straight current-carrying conductor.
- The figure shows such a conductor with length 2*a* carrying a current *I*.
- We will find $\vec{B}$ at a point a distance *x* from the conductor on its perpendicular bisector.




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# Magnetic Field of a Straight Current-Carrying Conductor (2 of 3)

- Since the direction of the magnetic field from all parts of the wire is the same, we can integrate the magnitude of the magnetic field and obtain:

$$B = \frac{\mu_0 I}{4\pi} \frac{2a}{x\sqrt{x^2 + a^2}}$$

- As the length of the wire approaches infinity, $a >> x$, and the distance $x$ may be replaced with $r$ to indicate this is a radius of a circle centered on the conductor:

**Magnetic field near a long, straight, current-carrying conductor**

$$B = \frac{\mu_0 I}{2\pi r}$$

Magnetic constant

Current

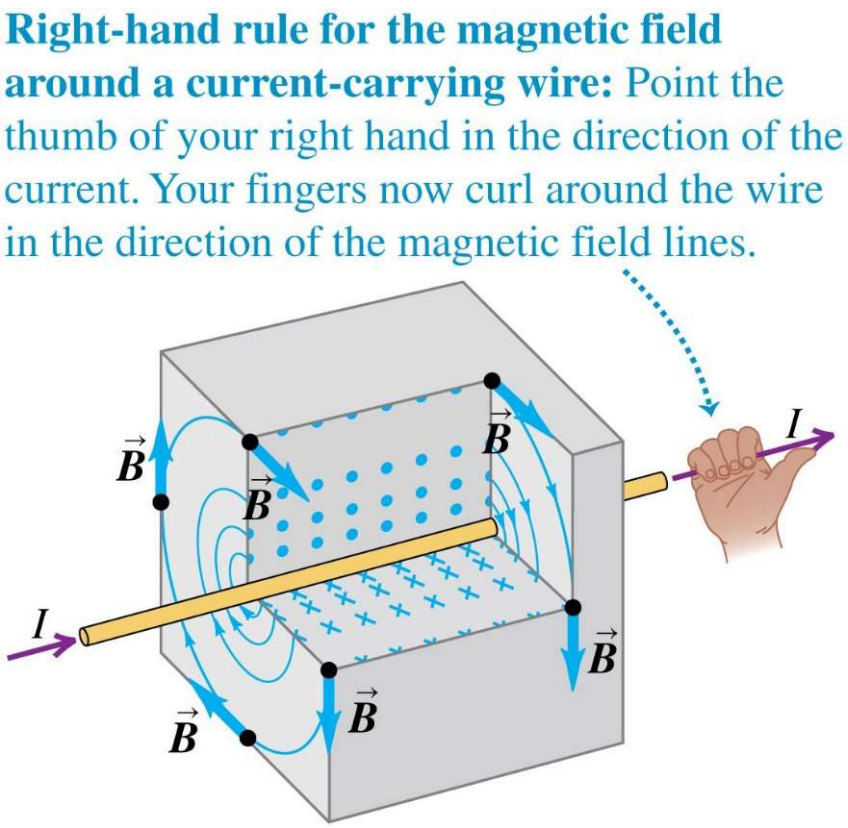
Distance from conductor

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# Magnetic Field of a Straight Current-Carrying Conductor (3 of 3)

- The field lines around a long, straight, current-carrying conductor are circles, with directions determined by the right-hand rule.




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# Magnetic Fields of Current-Carrying Wires

- Computer cables, or cables for audio-video equipment, create little or no magnetic field.
- This is because within each cable, closely spaced wires carry current in both directions along the length of the cable.
- The magnetic fields from these opposing currents cancel each other.

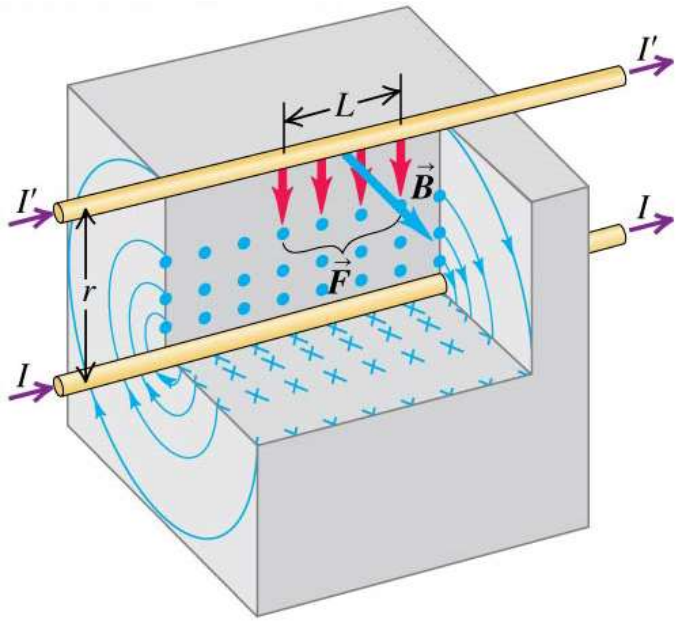


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# Force Between Parallel Conductors (1 of 2)

- The magnetic field of the lower wire exerts an **attractive** force on the upper wire.
- If the wires had currents in **opposite** directions, they would **repel** each other.

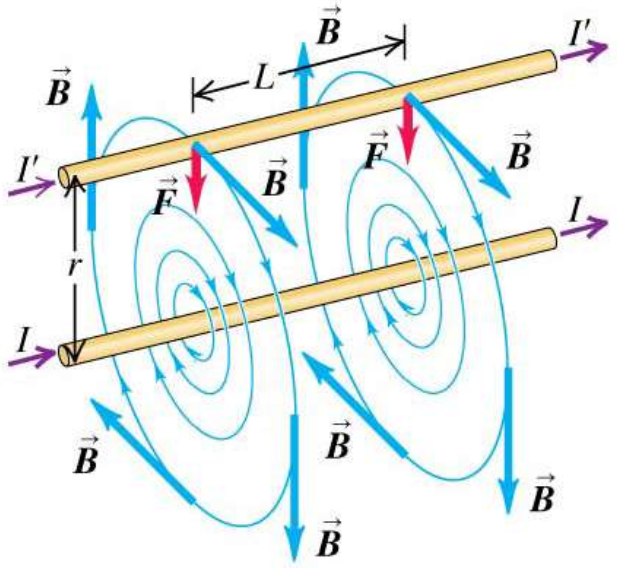



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# Force Between Parallel Conductors (2 of 2)

- The figure shows segments of two long, straight, parallel conductors separated by a distance *r* and carrying currents *I* and *I′* in the same direction.
- Each conductor lies in the magnetic field set up by the other, so each experiences a force.




**Magnetic force per unit length between two long, straight, parallel, current-carrying conductors**

$$\frac{F}{L} = \frac{\mu_0 I I'}{2\pi r}$$

- $\mu_0$: Magnetic constant
- $I$: Current in first conductor
- $I'$: Current in second conductor
- $r$: Distance between conductors

- Video Tutor Solution: Example 28.5

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# Definition of the Ampere

- The SI definition of the **ampere** is:

  **The ampere is a current of one coulomb per second, where the coulomb is defined in terms of the elementary charge *e*.**

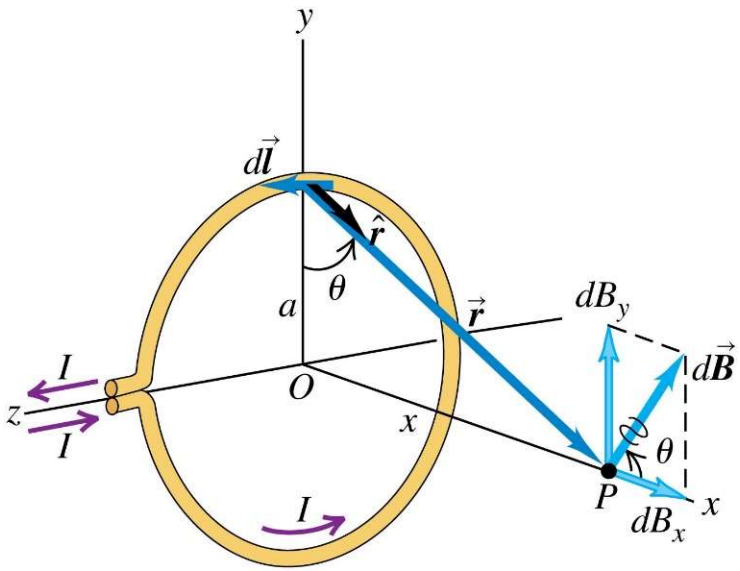
- This definition of the ampere is what leads us to choose the value of $4\pi \times 10^{-7}\,\mathrm{T \cdot m/A}$ for the magnetic constant, $\mu_0$.
- The coulomb is the quantity of electricity such that the elementary charge e is exactly $1.602176634 * 10^{-19}$ C transported in 1 second by a current of 1 ampere.

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# Magnetic Field of a Circular Current Loop (1 of 2)

- Shown is a circular conductor with radius *a* carrying a counterclockwise current *I*.
- We wish to calculate the magnetic field on the axis of the loop.

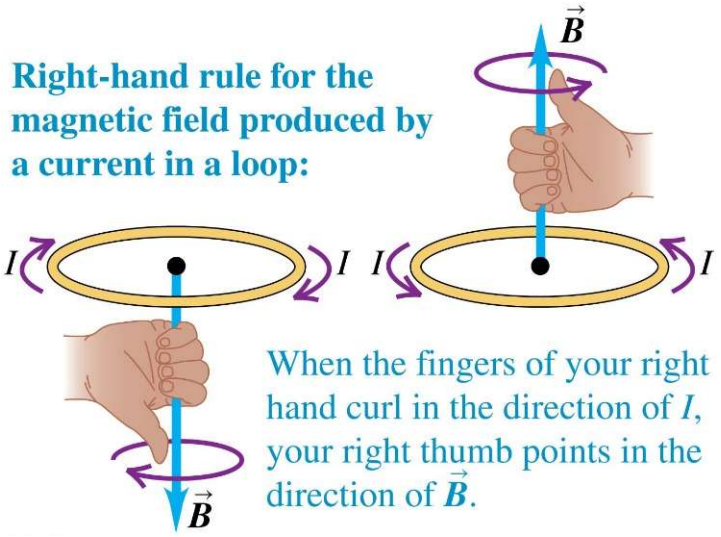



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# Magnetic Field of a Circular Current Loop (2 of 2)

- The magnetic field along the axis of a loop of radius *a* carrying a current *I* is given by the equation below.
- The direction is given by the right-hand rule shown.




**Magnetic field on axis of a circular current-carrying loop**

$$B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}$$

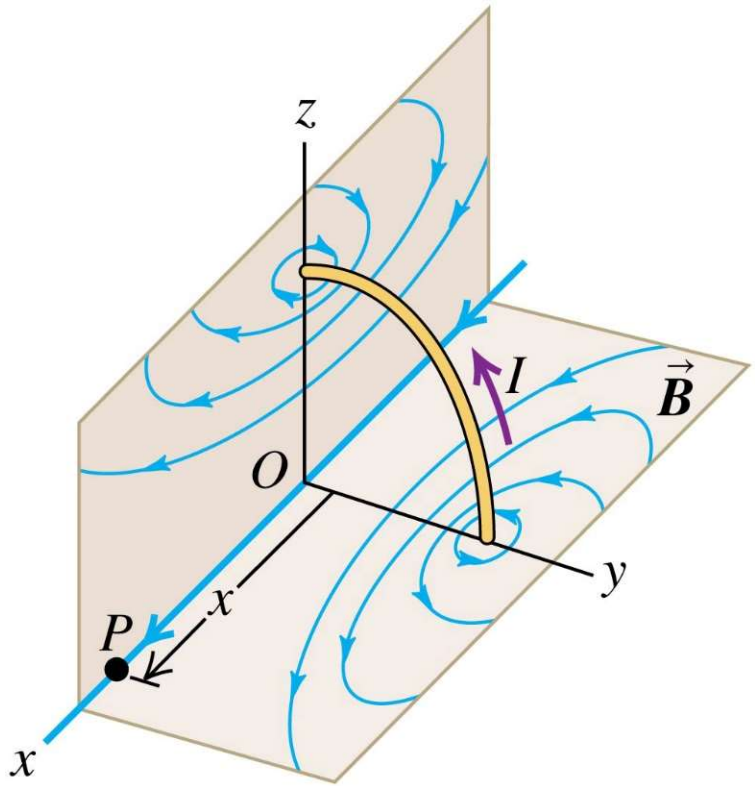
- $\mu_0$: Magnetic constant
- $I$: Current
- $a$: Radius of loop
- $x$: Distance along axis from center of loop to field point

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# Magnetic Field Lines of a Circular Current Loop

- The figure shows some of the magnetic field lines surrounding a circular current loop (magnetic dipole) in planes through the axis.
- The field lines for the circular current loop are closed curves that encircle the conductor; they are not circles, however.




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# Magnetic Fields for MRI

- MRI (magnetic resonance imaging) requires a magnetic field of about 1.5 T.
- In a typical MRI scan, the patient lies inside a coil that produces the intense field.
- The currents required are very high, so the coils are bathed in liquid helium at a temperature of 4.2 K to keep them from overheating.



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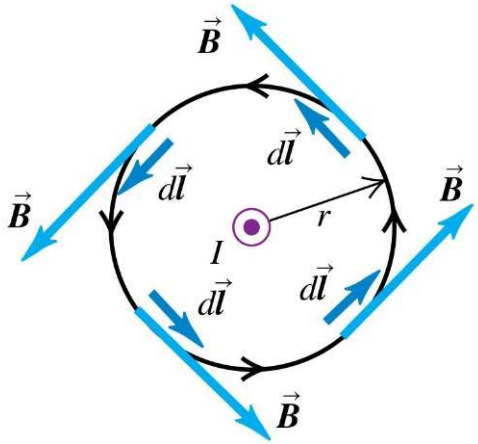
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# Ampere's Law (Special Case)

- **Ampere's law** relates electric current to the line integral around a closed path.
- Shown is the special case of a circular closed path centered on a long, straight conductor carrying current *I* out of the page.
- In this case the integral is simple (see figure to the right):

(a) Integration path is a circle centered on the conductor; integration goes around the circle counterclockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$

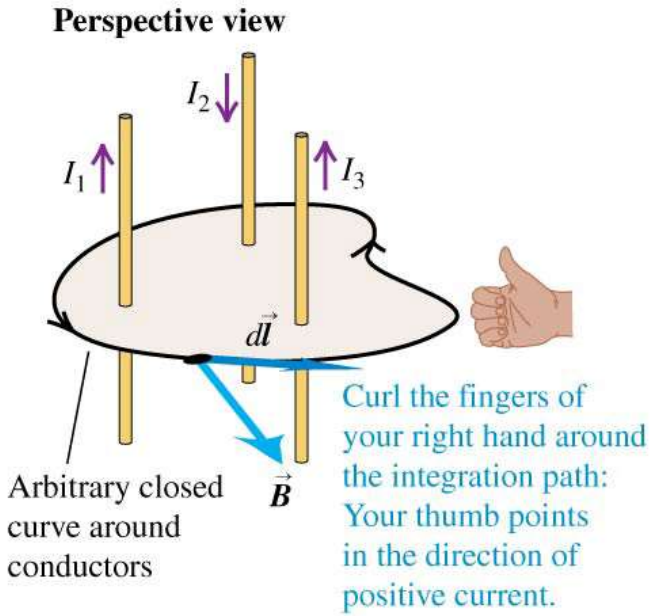



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# Ampere's Law (General Statement) (1 of 3)

- Suppose several long, straight conductors pass through the surface bounded by the integration path.
- Thus the line integral of the total magnetic field is proportional to the **algebraic sum** of the currents.

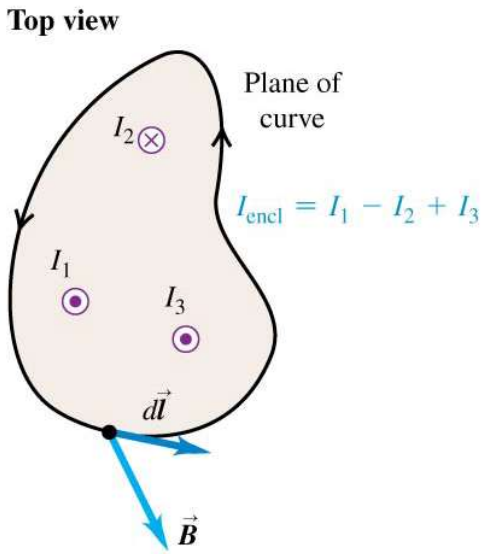



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# Ampere's Law (General Statement) (2 of 3)

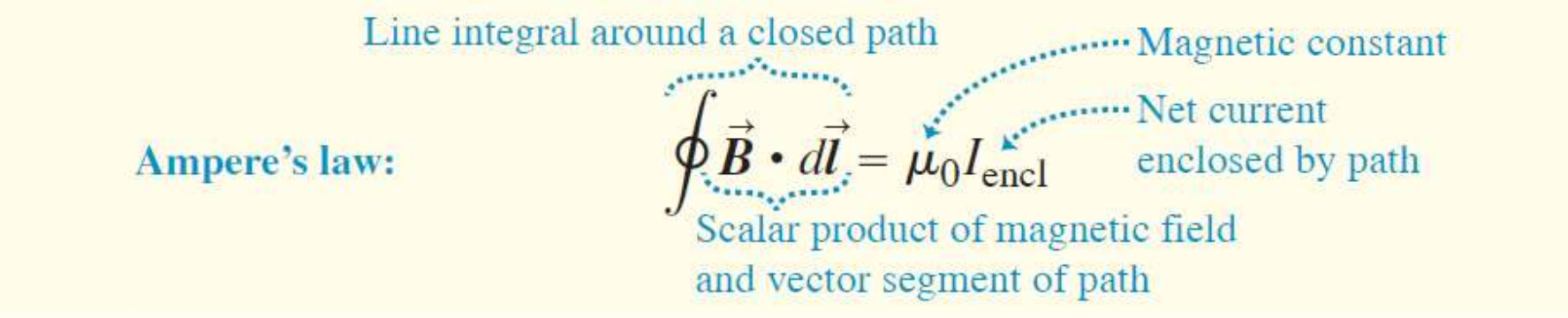
- For the general statement of Ampere's law, we can replace $I$ with $I_{encl}$, the algebraic sum of the currents enclosed or linked by the integration path, with the sum evaluated by using the right-hand sign rule.




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# Ampere's Law (General Statement) (3 of 3)



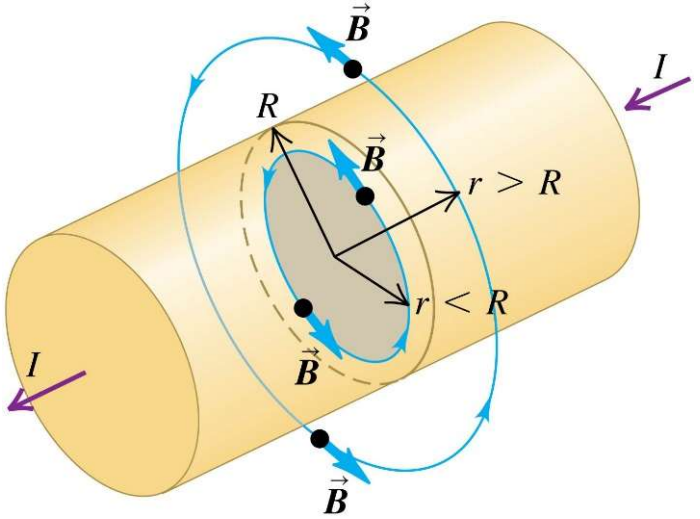

- This equation is valid for conductors and paths of any shape.
- If the integral around the closed path is zero, it **does not** necessarily mean that the magnetic field is everywhere along the path, only that the total current through an area bounded by the path is zero.

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# Field of a Long Cylindrical Conductor

- A cylindrical conductor with radius $R$ carries a current $I$.
- The current is uniformly distributed over the cross-sectional area of the conductor.
- To find the magnetic field, we apply Ampere's law and find:




$$B = \frac{\mu_0 I}{2\pi}\frac{r}{R^2} \text{ (inside the conductor, } r < R\text{)}$$

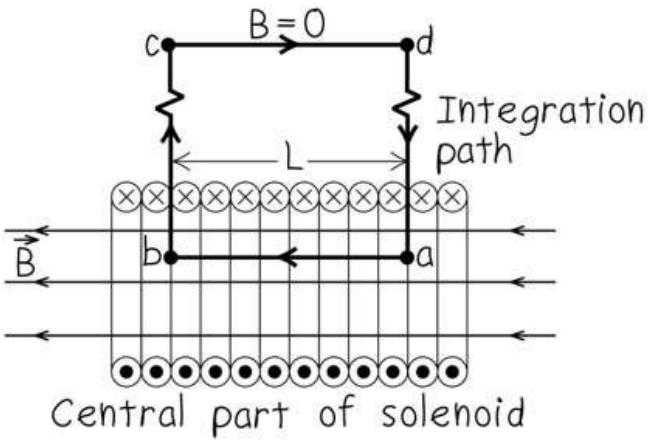
$$B = \frac{\mu_0 I}{2\pi r} \text{ (outside of the conductor, } r > R\text{)}$$

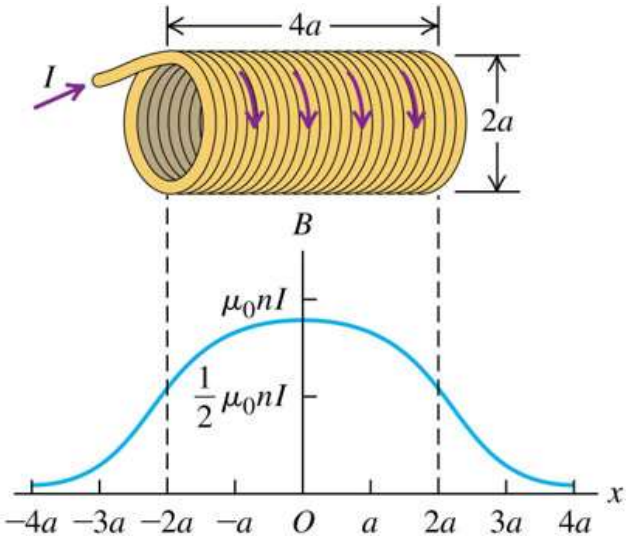
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# Field of a Solenoid

- A **solenoid** consists of a helical winding of wire on a cylinder.
- Follow Example 28.9 using the figures below.



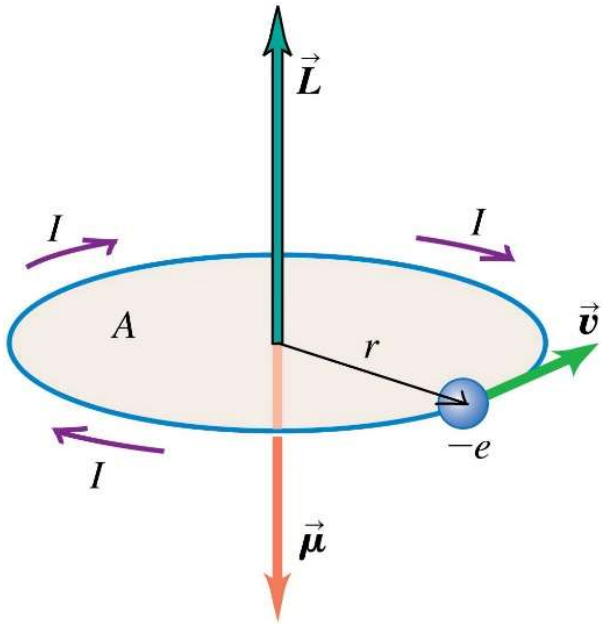

- Video Tutor Solution: Example 28.9

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# The Bohr Magneton (1 of 2)

- An electron moving with speed *v* in a circular orbit of radius *r* has an angular momentum $\vec{L}$ and an oppositely directed orbital magnetic dipole moment $\vec{\mu}$.
- It also has a spin angular momentum and an oppositely directed spin magnetic dipole moment.




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# The Bohr Magneton (2 of 2)

- As an electron orbits the nucleus of an atom, its magnetic dipole moment has a magnitude proportional to its orbital angular momentum.

$$\mu = \frac{e}{2m}L$$

- Atomic angular momentum is **quantized**; its component in a particular direction is always an integer multiple of $\frac{h}{2\pi}$, where $h = 6.626 \times 10^{-34}$ J·s is Planck's constant.
- This means there is a fundamental unit of magnetic dipole moment, which is called the **Bohr magneton**,

$$\mu_B = eh/4\pi m = 9.274 \times 10^{-24} \text{ J/T}.$$

- Electron spin also leads to a magnetic moment, which is about equal to one Bohr magneton.

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# Paramagnetism and Diamagnetism

- When an external magnetic field permeates a **paramagnetic** material, the result is that the magnetic field at any point is *greater* by a dimensionless factor $K_m$, called the relative permeability of the material, than it would be if the material were replaced by vacuum.
- If an external magnetic field permeates a **diamagnetic** material, the result is a magnetic field that is slightly *less* than it would be if the material were replaced by vacuum.
- The amount by which the relative permeability differs from unity is called the magnetic susceptibility, denoted by $\chi_m$:

$$\chi_m = K_m - 1$$

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# Magnetic Susceptibilities of Certain Materials

**Table 28.1** Magnetic Susceptibilities of Paramagnetic and Diamagnetic Materials at $T = 20°C$

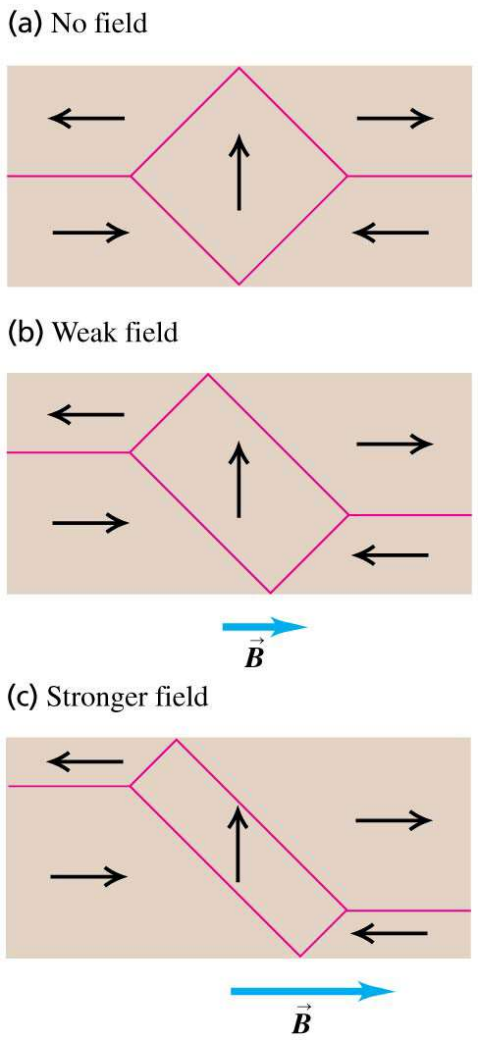
| **Material** | $\chi_m = K_m - 1 (\times 10^{-5})$ |
|---|---|
| **Paramagnetic** | |
| Iron ammonium alum | 66 |
| Uranium | 40 |
| Platinum | 26 |
| Aluminum | 2.2 |
| Sodium | 0.72 |
| Oxygen gas | 0.19 |
| **Diamagnetic** | |
| Bismuth | −16.6 |
| Mercury | −2.9 |
| Silver | −2.6 |
| Carbon (diamond) | −2.1 |
| Lead | −1.8 |
| Sodium chloride | −1.4 |
| Copper | −1.0 |

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# Ferromagnetism

- In **ferromagnetic materials** (such as iron), atomic magnetic moments tend to line up parallel to each other in regions called magnetic domains.
- When there is no externally applied field, the domain magnetizations are randomly oriented.
- When an external magnetic field is present, the domain boundaries shift; the domains that are magnetized in the field direction grow, and those that are magnetized in other directions shrink.




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# End of Chapter

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